17-W1　两种轨道方向的诱导磁矩

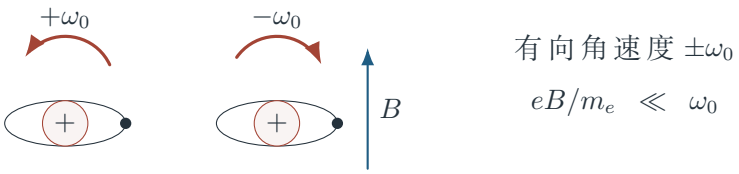


图 17-13　题图　两条轨道的运动方向相反，取同一个 z 轴

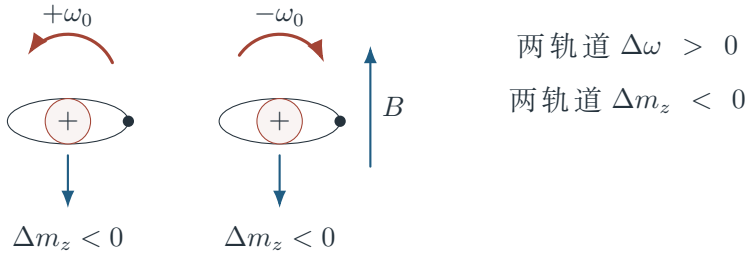
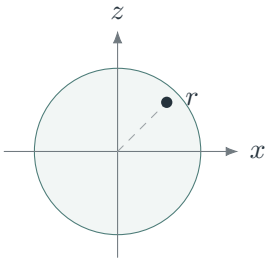


图 17-14　解答分析图　转速大小的变化相反，诱导磁矩的方向相同

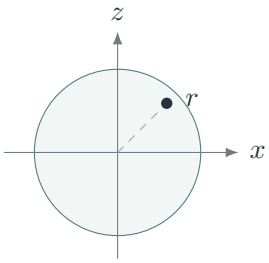
17-W2　各向同性电子分布的抗磁率



三维各向同性分布

$$\langle r_j^2 \rangle \;=\; R^2$$

图 17-15　题图　图示一个电子的位置，统计平均包含全部方向

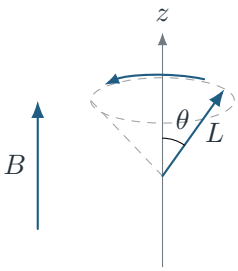


$$\langle x^2 + y^2 \rangle \;=\; 2R^2/3$$

$$\chi_m \;=\; -\mu_0 n_a Z e^2 R^2/(6m_e)$$

图 17-16　解答分析图　抗磁率中使用三维均方半径

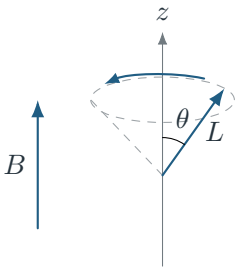
17-W3　磁力矩产生的进动



$m = \gamma L,\ \gamma < 0$

恒定场，无耗散

图 17-17　题图　角动量的横向分量绕 z 轴转动



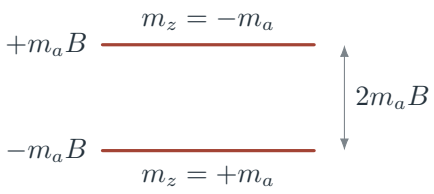
$\Omega = -\gamma B > 0$

$\theta = \theta_0$

$E = -m \cdot B$ 不变

图 17-18　解答分析图　保守力矩产生进动，夹角保持不变

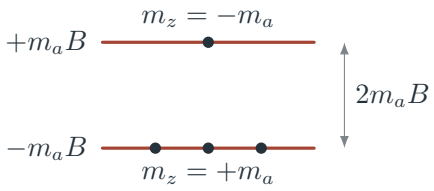
17-W4　两能级顺磁体的磁化与热容



独立两能级磁矩

$$\xi = m_aB/(k_BT)$$

图 17-19　题图　磁矩顺场时能量较低



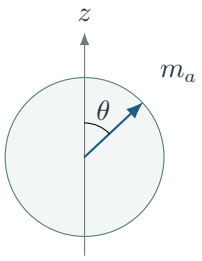
$$M = nm_a \tanh \xi$$

$$u = -BM$$

$$c_B = nk_B\xi^2 \operatorname{sech}^2 \xi$$

图 17-20　解答分析图　温度改变能级布居，改变磁化和磁性热容

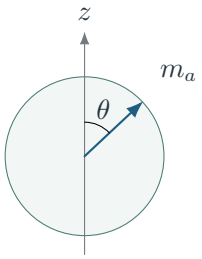
17-W5　连续取向的顺磁平均



连续球面取向

权重 $e^{\xi\cos\theta}\sin\theta$

图 17-21　题图　球面面积元含正弦因子，不能只对极角等权平均



$\langle\cos\theta\rangle = \mathcal{L}(\xi)$

弱场：$\mathcal{L} \simeq \xi/3$

图 17-22　解答分析图　弱场系数 1/3 来自零场的余弦平方平均

17-W6　居里–外斯直线与退磁修正

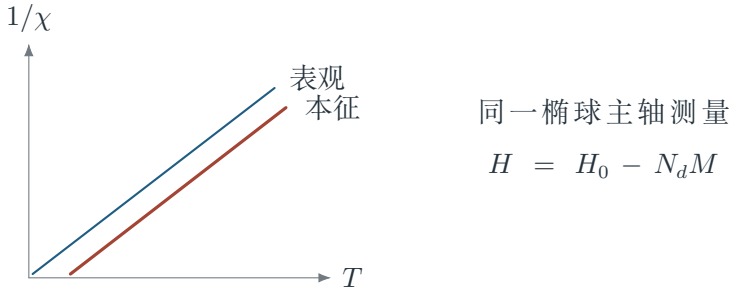


图 17-23　题图　两条逆磁化率直线的斜率相同

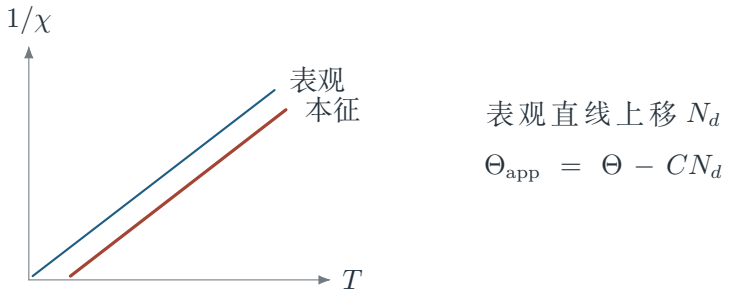
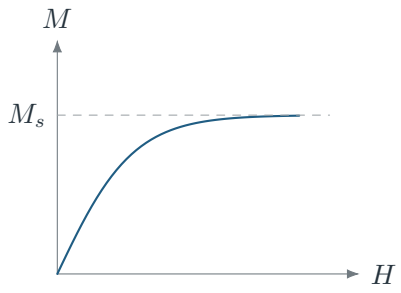


图 17-24　解答分析图　退磁修正改变温度截距，图示正退磁因子情形

17-W7　饱和磁化和小信号磁导率

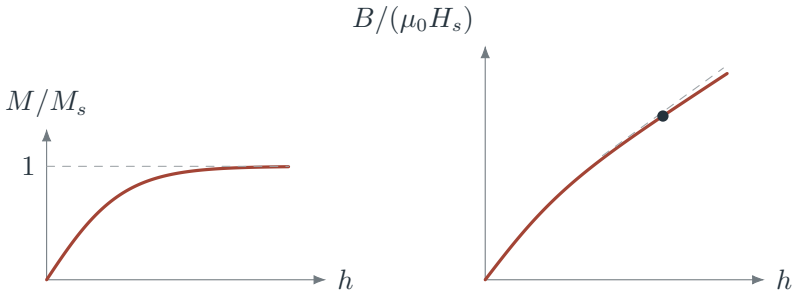


$$M \;=\; M_s \tanh(H/H_s)$$

可逆模型，无回线

图 17-25　题图　M 趋于饱和，B 仍有真空场贡献

饱和后仍有非零斜率

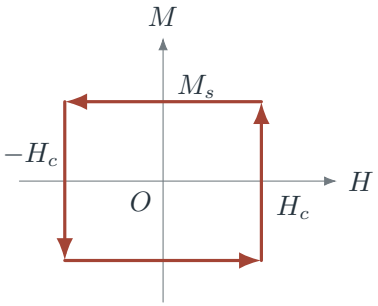


$$h = H/H_s;\ \text{示意取 } M_s/H_s = 1$$

$$\mu_{\text{diff}} = \mu_0[1 + (M_s/H_s)\,\operatorname{sech}^2 h] \;\longrightarrow\; \mu_0$$

图 17-26　解答分析图　分别画 M 与 B 的饱和响应，B 曲线局部切线保留真空磁导率项

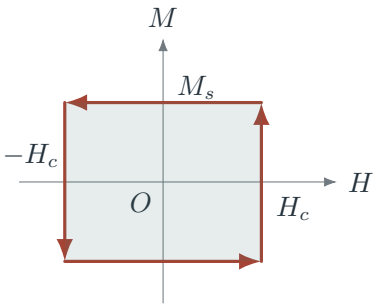
17-W8　矩形磁滞回线的损耗



M–H 矩形回线

循环越过两个阈值

图 17-27　题图　按先增场后减场的循环，两个跃迁均耗能

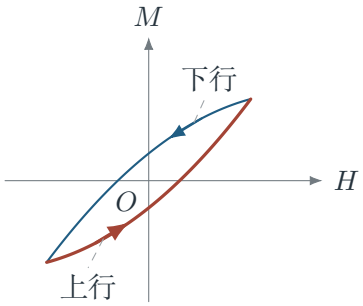


几何面积 $4H_cM_s$

能量密度再乘 μ_0

图 17-28　解答分析图　染色面积表示 M–H 几何面积

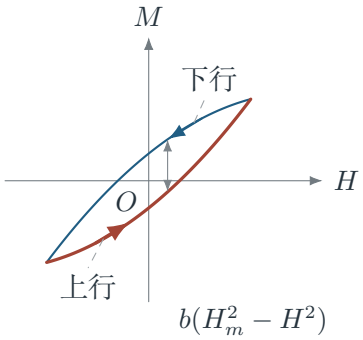
17-W9　瑞利次回线的面积



对称幅值 H_m

$a > 0,\ b > 0$

图 17-29　题图　上行支路在下，下行支路在上

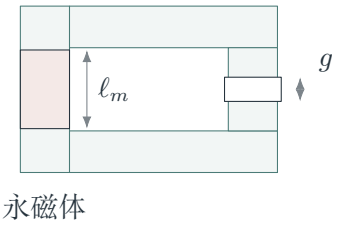


$w_h = 4\mu_0 b H_m^3/3$

$M_r = bH_m^2/2$

图 17-30　解答分析图　积分两支路之差，线性可逆项相消

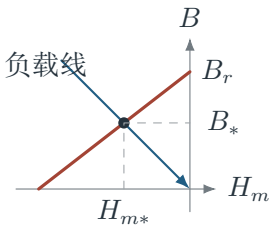
17-W10　永磁气隙的最佳工作点



无自由绕组

等截面，磁轭磁阻忽略

图 17-31　题图　气隙使永磁体承受反向内部场

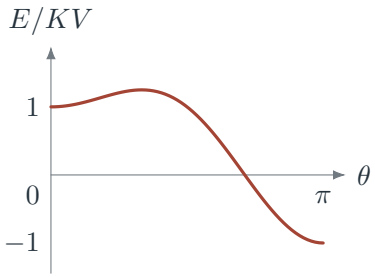


$g_*/\ell_m = 1/\mu_{\mathrm{rec}}$

需未越过退磁拐点

图 17-32　解答分析图　负载线与可逆退磁支路共同确定工作点

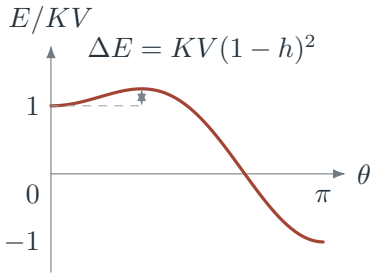
17-W11　反向场中的单畴能垒



图取 $h = 1/2$

$E/KV = \sin^2\theta + 2h\cos\theta$

图 17-33　题图　反向场使反向磁化的能量较低，初态仍有能垒



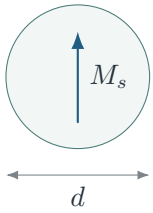
鞍点 $\cos\theta_b = h$

初态 $\theta = 0$ 亚稳

$h = 1$ 时失稳

图 17-34　解答分析图　反转能垒从初态能量量起

17-W12　阻塞温度与测量时间

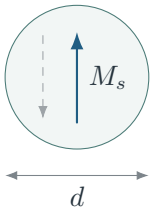


$$V = \pi d^3/6$$

$$\tau = \tau_0 e^{KV/(k_B T)}$$

图 17-35　题图　单畴颗粒仍可整体反转磁矩

热激活反转

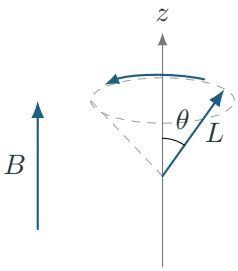


$\tau = t_m$ 定义阻塞温度

更短 t_m：更高 T_B

图 17-36　解答分析图　是否观测到剩磁取决于反转时间与测量时间之比

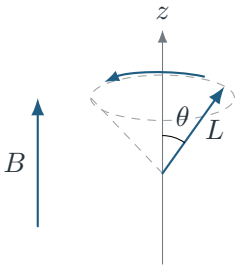
17-T1　轨道与自旋的旋磁比



轨道：$|\gamma| = e/(2m_e)$

自旋：$|\gamma| = e/m_e$

图 17-37　题图　两种角动量用各自的旋磁比描述



题给磁矩大小相等

自旋进动频率为轨道的 2 倍

图 17-55　解答分析图　相同磁矩大小不意味着相同进动频率

17-T2　固定半径模型的两条轨道

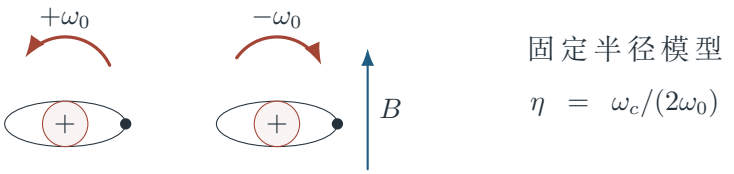


图 17-38　题图　保留两条轨道的精确代数根

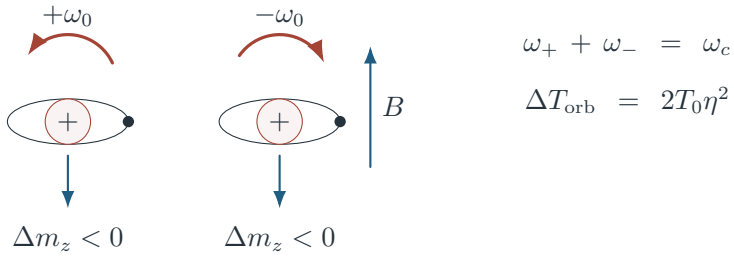


图 17-56　解答分析图　模型内总磁矩线性，总动能增量为二阶

17-T3　顺磁磁化的温度比

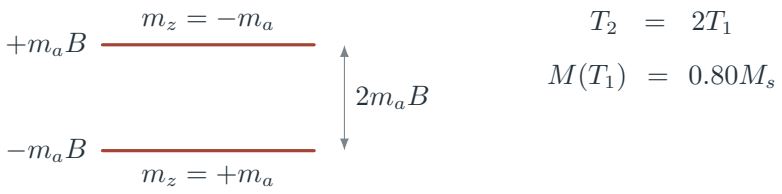


图 17-39　题图　场和能级差相同，只有温度改变

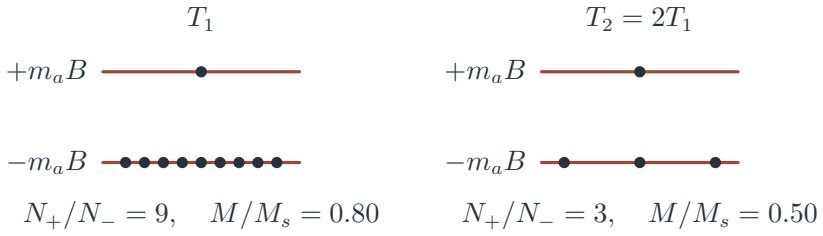
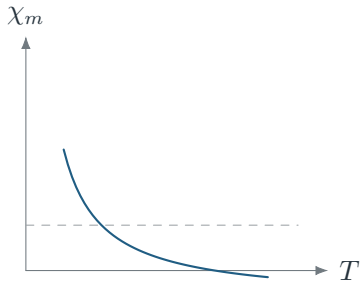


图 17-57　解答分析图　同一磁场下，升温将低能级与高能级布居比从 9 变为 3

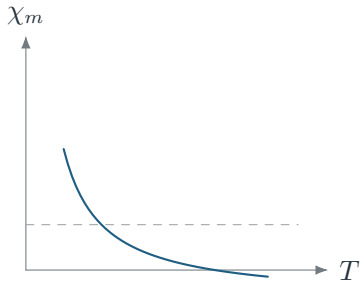
17-T4　顺磁与抗磁贡献的抵消



$$\chi_m = C/T - D$$

$$\nabla H_0^2 \parallel +z$$

图 17-40　题图　顺磁贡献随温度升高而减小

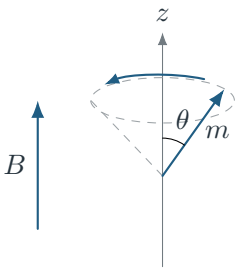


$$T_* = C/D$$

低温吸向强场，高温趋弱场

图 17-58　解答分析图　图中水平虚线是零磁化率，交点处为宏观抵消

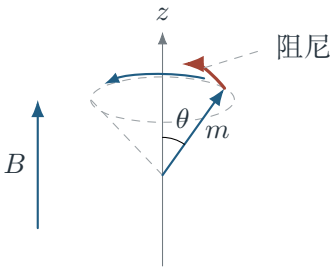
17-T5　带阻尼的磁矩取向



进动项与阻尼项

$$\alpha, \gamma \ > \ 0$$

图 17-41　题图　阻尼使磁矩沿球面向 z 轴靠拢



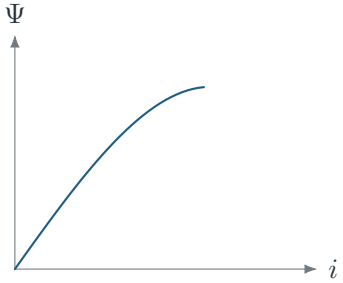
$$\dot{\phi} \ = \ \gamma B$$

$$\dot{\theta} \ = \ -\alpha\gamma B \sin\theta$$

$$\dot{E} \ \leq \ 0$$

图 17-59　解答分析图　转角随时间减小，磁矩大小不变

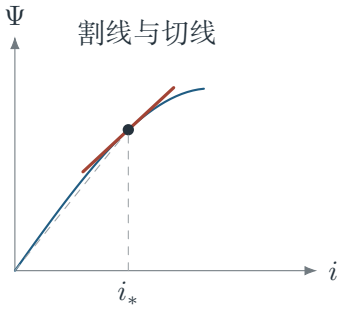
17-T6　非线性铁芯的微分电感



$$B = \mu_i H - \beta H^3$$

仅在正微分磁导率区使用

图 17-42　题图　非线性磁链曲线按题给材料关系绘制

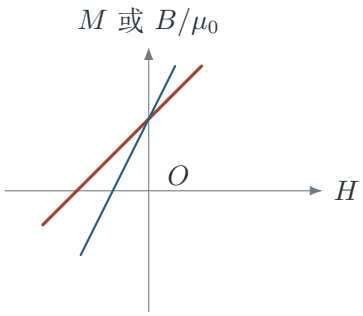


比例电感：Ψ/i

微分电感：$\mathrm{d}\Psi/\mathrm{d}i$

图 17-60　解答分析图　沿切线的斜率决定电压与电流变化率

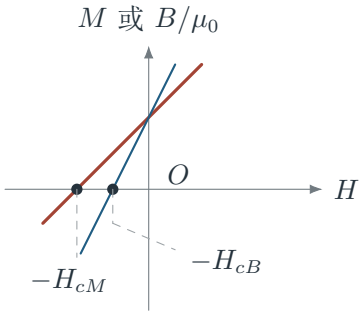
17-T7　两种矫顽力



下降支路 $M = M_r + \chi_d H$

红：M，蓝：B/μ_0

图 17-43　题图　两条支路在 H=0 相交，但零点不同

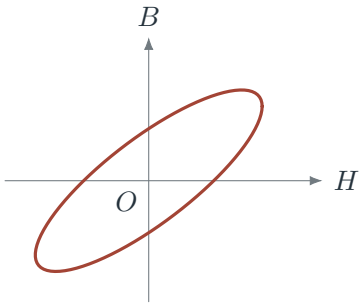


$H_{cM} > H_{cB}$

$B = 0$ 时仍有 $M > 0$

图 17-61　解答分析图　图示磁化支路斜率为 1 的比例，两个矫顽场分别标出

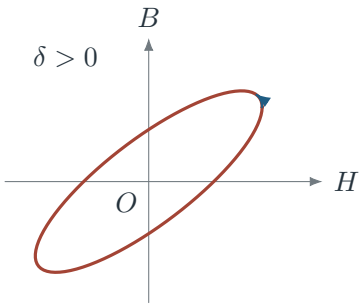
17-T8　椭圆动态回线



$$H = H_p \cos \omega t$$

$$B = B_p \cos(\omega t - \delta)$$

图 17-44　题图　B 滞后 H，形成椭圆回线

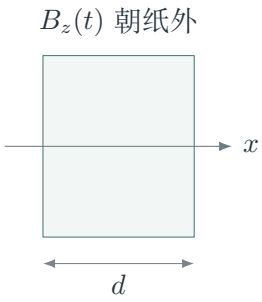


$$w = \pi H_p B_p \sin \delta$$

按时间方向积分为正

图 17-62　解答分析图　图取相位延迟 35 度，右端向上走，输入功为正

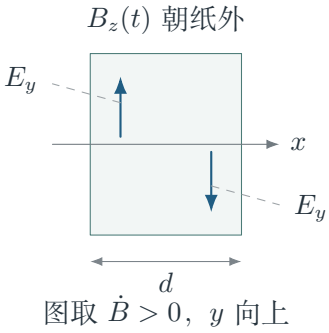
17-T9　薄片涡流的厚度标度



片面为 yz 平面

无输运电流

图 17-45　题图　厚度方向 x 垂直片面，B 沿片面 z 方向



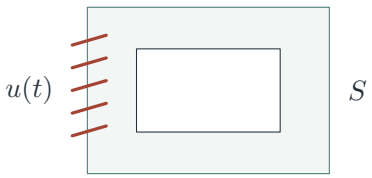
图取 $\dot{B} > 0$，y 向上

$$E_y = -x\dot{B}$$

$$\langle P_e \rangle / V \propto f^2 d^2$$

图 17-63　解答分析图　两侧感应电场反向，有限片内电流经边缘闭合

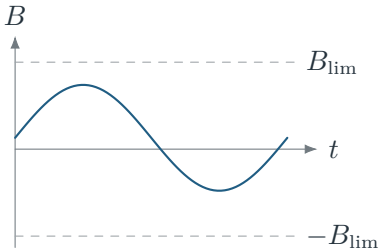
17-T10　电压驱动与饱和时间



理想电压驱动

$$u = NS\dot{B}$$

图 17-46　题图　法拉第定律直接约束磁通变化



偏置 B_0 占用磁通余量

直流电压使 B 持续上升

图 17-64　解答分析图　图示带正偏置的交流磁感应及规定界限

17-T11　软磁样品断电后的剩磁



$H_0 = 0$

撤去外场：$H_0 = 0$

$M = M_r + \chi_{\mathrm{rec}} H$

图 17-47　题图　样品几何决定撤去外场后的反向内部场



$H_0 = 0$

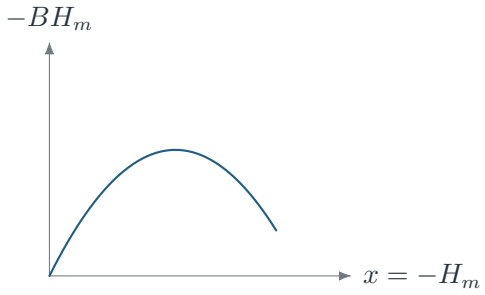
$H = -N_d M$

$M = M_r/(1 + N_d \chi_{\mathrm{rec}})$

闭合环可保留剩磁

图 17-65　解答分析图　图为正磁化，退磁场与磁化反向

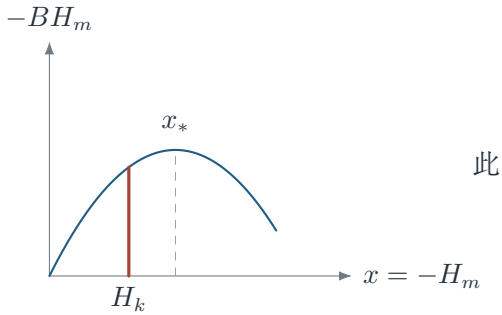
17-T12　受退磁拐点限制的磁能积



允许 $0 \le x \le H_k$

$B = B_r - \mu_0\mu_{\mathrm{rec}}x$

图 17-48　题图　磁能积是一条开口向下的抛物线

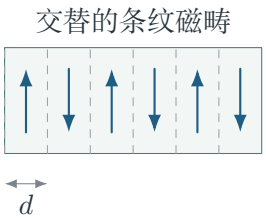


图示 $H_k < x_*$

此时最优点取允许边界

图 17-66　解答分析图　拐点限制可能使无约束顶点无法使用

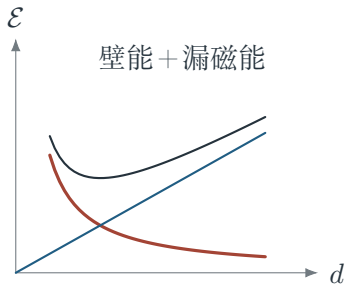
17-T13　畴壁能与条纹宽度



壁能随 $1/d$ 减小

漏磁能随 d 增加

图 17-49　题图　条纹宽度在两个相反能量趋势之间选定

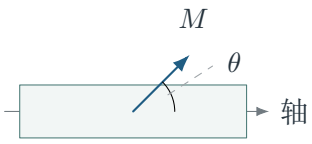


$d_* \propto \sqrt{t}$

最优点两能量项相等

图 17-67　解答分析图　总能量的最小值决定模型内的畴宽

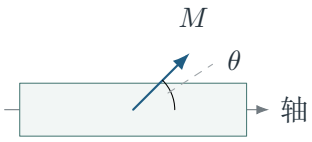
17-T14　磁致伸缩与倍频



$$\epsilon \propto \cos^2\theta - 1/3$$

磁化方向决定轴向应变

图 17-50　题图　应变沿图中水平轴测量



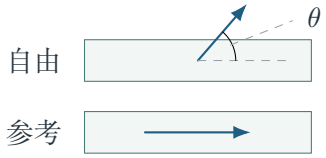
180 度反转：应变不变

垂直轴偏置：首项为二倍频

45 度偏置：首阶灵敏度最大

图 17-68　解答分析图　应变对磁化反向为偶函数

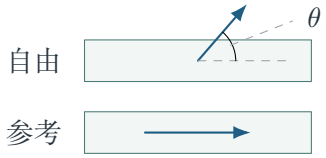
17-T15　磁电阻传感器的工作点



恒流 I_r 读出

$R = R_P + \Delta R(1-\cos\theta)/2$

图 17-51　题图　两磁性层的相对方向进入题给电阻关系



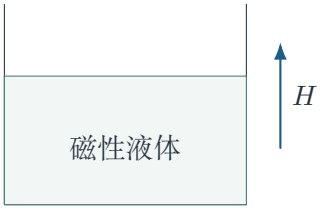
$\mathrm{d}U/\mathrm{d}\theta \propto \sin\theta$

90 度偏置有一倍频信号

共线偏置先有二倍频

图 17-69　解答分析图　最大首阶灵敏度位于相对磁化垂直时

17-T16　理想磁性液体的静压



给定静压关系

$$\mathrm{d}p = \mu_0 M\,\mathrm{d}H$$

图 17-52　题图　用同一低场压力基准比较液面高度



$$\Delta p = nk_B T \ln(\sinh\xi/\xi)$$

$$\rho_f g_0 \Delta z = \Delta p$$

图 17-70　解答分析图　高度示意按题给静压模型，不包含任意界面磁应力

17-T17　斜场反转与热稳定性

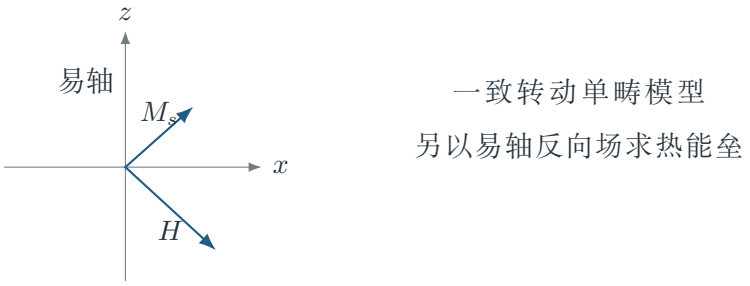


图 17-53　题图　斜场问题与沿易轴的热写入问题分别求解

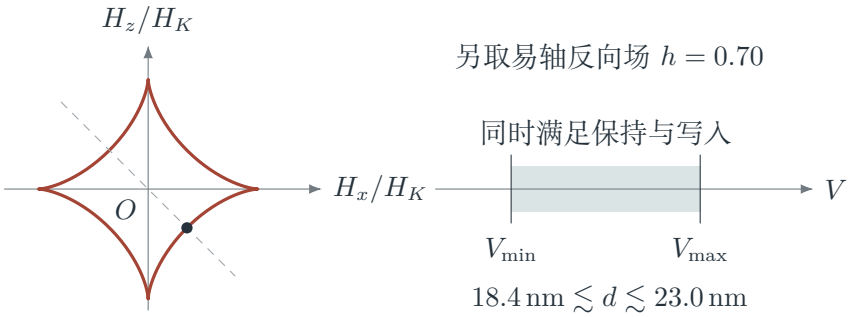
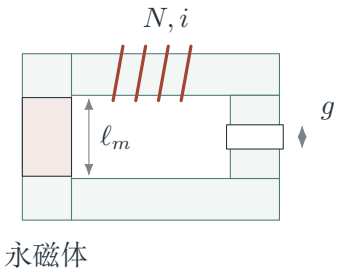


图 17-71　解答分析图　星形曲线与易轴场下的热稳定体积区间分别对应两问的约束

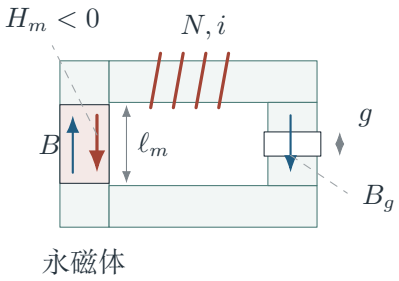
17-T18　温度变化下的永磁磁路



B_r, H_k 随温度改变

反向脉冲 $Ni = -\mathcal{I}$

图 17-54　题图　绕组安匝正向帮助原磁化，负向削弱它



先算内部反向场 H_m

同时满足拐点与 B 符号限制

开隙使反向内部场增强

图 17-72　解答分析图　温度、气隙和电流共同移动工作点